

Dynamic Visual SLAM with Temporal Semantic Priors and Disparity-Gated Confidence

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Abstract. Visual Simultaneous Localization and Mapping (V-SLAM) in dynamic environments remains a significant challenge, as moving objects violate the static-world assumption and induce erroneous feature associations. Although semantic-assisted methods provide a pathway for dynamic feature rejection, they encounter a critical trade-off: computationally intensive segmentation networks ensure robustness at the expense of real-time performance, whereas lightweight detectors often lack the fidelity required for precise dynamic filtering. To address this, we propose TP-DSLAM, a robust stereo SLAM framework that integrates Temporal-Semantic Priors with Geometric-Guided Modulation to achieve a favorable balance between efficiency and accuracy. Our method introduces a Temporal-Latent Semantic Propagation (TLSP) strategy that executes object detection at sparse intervals and propagates high-confidence priors to intermediate frames, thereby reducing computational overhead. Furthermore, inspired by the inherent property that near-field objects own reliable stereo geometric evidence while far-field regions suffer from low disparity confidence, we introduce Disparity-Gated Confidence Modulation (DGCM). DGCM fuses binocular geometric constraints with semantic likelihood into a joint dynamic posterior: it strictly suppresses dynamic features in the near-field where geometric confidence is high, while smoothly transitioning to rely on semantic priors in the far-field. This integration refines coarse semantic masks into fine-grained dynamic confidence maps. Extensive experiments demonstrate that our method reduces the Absolute Trajectory Error (ATE) by 3%-60% compared to ORB-SLAM3 in highly dynamic sequences while maintaining real-time performance on edge devices.

Keywords: Visual SLAM · Dynamic Environments · Temporal-Semantic Fusion · Disparity-Gated Modulation · Stereo Disparity.

1 Introduction

Visual Simultaneous Localization and Mapping (Visual SLAM) has emerged as a cornerstone technology for mobile robotics and autonomous navigation. While state-of-the-art frameworks, such as ORB-SLAM3, have demonstrated remarkable robustness in static environments, they fundamentally rely on the static-world assumption[4]. In real-world dynamic scenarios—such as crowded corridors or urban traffic—moving objects introduce incorrect feature associations. This

violation of static constraints often leads to degraded pose estimation and map corruption, posing a critical challenge for long-term autonomy[2, 19].

To handle dynamic objects, existing methods broadly split into two paradigms: geometry-based and semantics-assisted approaches. Geometry-based methods exploit optical flow, motion consistency, or epipolar constraints to filter dynamic features[8, 3]. While computationally lightweight, they often fail in textureless regions or under fast motion, struggling to reliably separate dynamic from static elements. Semantics-assisted methods integrate deep learning (e.g., Mask R-CNN or YOLO) to identify and reject dynamic instances, substantially improving robustness[18, 2]. However, they face a critical trade-off: heavy segmentation networks ensure accuracy at the cost of real-time performance, whereas lightweight detectors sacrifice fidelity, producing coarse boundaries that hurt filtering precision[11, 16]. Moreover, most semantics-assisted systems adopt hard-rejection strategies—binary removal of features within dynamic masks—which ignore predictive uncertainty and frequently discard static background features near object boundaries[19, 14].

Parallel to these advances, a third line of work has emerged with a fundamentally different goal: high-fidelity dense reconstruction using neural representations, most recently 3D Gaussian Splatting (3DGS)-based SLAM[9, 17]. These methods achieve photorealistic rendering but prioritize visual quality over localization efficiency. Their heavy GPU dependence makes them ill-suited for real-time pose tracking on resource-constrained edge devices[6, 5]. Motivated by these limitations, this paper presents TP-DSLAM, a robust stereo Visual SLAM framework that achieves a favorable balance between efficiency and accuracy in dynamic environments. Unlike conventional approaches that rely on heavy per-frame inference or binary masking, our method introduces a Temporal-Semantic Fusion strategy and a Soft-Weighted Optimization mechanism. Specifically, we develop a Temporal-Latent Semantic Propagation (TLSP) strategy to reduce computational overhead by reusing semantic priors across frames.

Furthermore, we formulate a Disparity-Gated Confidence Modulation (DGCM) mechanism based on a clear physical interpretation: binocular disparity directly correlates with the reliability of geometric dynamic evidence. DGCM constructs a depth-aware manifold that acts as a gate; for near-field objects with significant motion parallax, it strongly enforces dynamic suppression. Conversely, in the far-field where geometric evidence diminishes, it smoothly transitions to rely on semantic priors. This joint dynamic posterior enables fine-grained suppression of dynamic objects while preserving static scene structure.

The main contributions of this work are summarized as follows:

- Temporal Propagation (TLSP)**: We introduce TLSP, which executes object detection only every K frames and propagates semantic priors via pose-guided warping with dilated potential regions. This strategy reduces per-frame inference overhead by 5-10%, and ablation shows that removing TLSP degrades ATE from 0.026 m to 0.031 m on the TUM w_half sequence.

- Soft-Weighted Modulation (DGCM)**: DGCM fuses semantic likelihood with a disparity-based geometric gate to create a joint dynamic posterior. By

modeling the depth-motion relationship, this soft-weighted approach effectively suppresses dynamic features, significantly outperforming hard rejection and preventing major accuracy degradation.

•**Lightweight Framework:** We propose a dynamic SLAM pipeline built upon ORB-SLAM3, specifically tailored for edge devices, ensuring robust operation.

2 Related Works

2.1 Temporal Semantic Propagation in Dynamic SLAM

Existing semantic SLAM methods alleviate the heavy cost of per-frame inference by performing semantic extraction only on keyframes and propagating results to intermediate frames. Representative approaches such as [12] adopt mesh-flow and local affine transformation for semantic propagation, which purely relies on 2D optical flow consistency without considering full 3D camera rigid motion.

By contrast, the proposed TLSP module constructs geometry-guided warping using the global $SE(3)$ pose estimated by visual odometry. It propagates semantic priors according to real camera motion, achieving better geometric coupling with SLAM frontend and backend optimization. Furthermore, a dilated potential region strategy is introduced to compensate for dynamic motion uncertainty, effectively improving robustness in highly dynamic scenes.

2.2 Geometry-Semantic Fusion via Disparity Constraints

In dynamic SLAM research, the fusion of geometric and semantic cues serves as the key to improving localization robustness. Early methods such as [1] rely purely on geometric constraints including optical flow and epipolar geometry to eliminate dynamic features. Nevertheless, they tend to fail in textureless regions or under rapid camera motion. Subsequently, semantic SLAM methods represented by [20] jointly solve object detection and geometric features via tightly coupled optimization, which remarkably enhances mapping accuracy and consistency in dynamic environments.

However, existing fusion methods generally integrate semantic confidence and geometric constraints with fixed static weighting, ignoring the reliability variance of geometric evidence at different depth ranges. To address this limitation, this paper proposes the Disparity-Gated Confidence Modulation (DGCM) mechanism, which leverages stereo disparity as a depth-aware geometric prior. The proposed mechanism strictly suppresses dynamic features in near-field regions with high geometric credibility, while smoothly transitioning to rely on semantic priors in far-field areas with low geometric reliability, thereby achieving adaptive dynamic fusion of geometric and semantic cues.

3 Methodology

3.1 System Overview

The overall architecture of the proposed TP-DSLAM is illustrated in Fig. 1. Our system adopts a tracking-first paradigm inherited from ORB-SLAM3. The whole framework takes stereo visual and measurement data as input, and progressively completes sparse semantic propagation, geometric-semantic confidence fusion, and manifold-based robust pose refinement for dynamic scenarios.

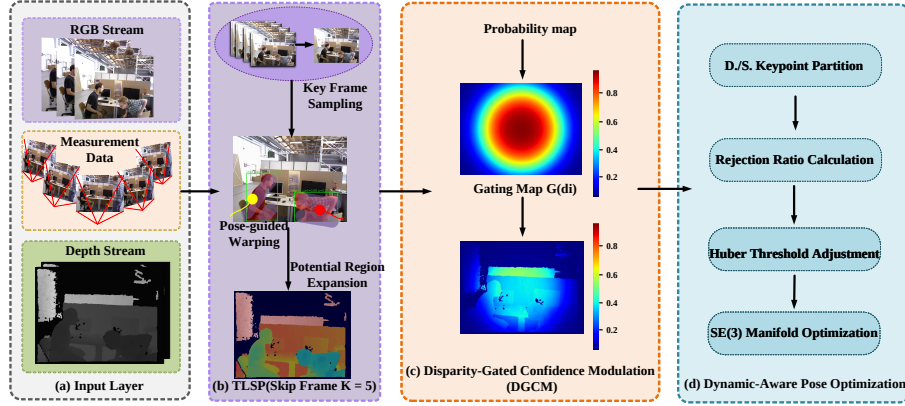


Fig. 1. Overall Architecture of TP-DSLAM.

Given a stream of stereo image pairs $\{I_t, D_t\}$ together with sensor measurement data, the system executes a complete forward pipeline layer by layer, and the data flow and functional logic of each module are described as follows:

(a) Input Layer. This module synchronizes and outputs aligned continuous stereo RGB stream, depth stream and auxiliary measurement data, which serve as the raw input for subsequent semantic propagation and geometric constraint calculation.

(b) Temporal-Latent Semantic Propagation (TLSP). To balance computational efficiency and localization accuracy, TLSP adopts a skip-frame mechanism with interval $K = 5$. It first implements key frame sampling, then completes semantic prior propagation via pose-guided warping relying on the initial pose estimated by the front-end visual odometry. After that, it expands potential dynamic regions to compensate for motion uncertainty, and finally outputs warped image pairs and coarse dynamic potential region priors to the subsequent DGCM module.

(c) Disparity-Gated Confidence Modulation (DGCM). Taking the semantic prior and stereo disparity information from TLSP as input, DGCM constructs pixel-level dynamic probability maps, and further generates a disparity-aware gating map by fusing binocular geometric credibility. This module refines

coarse semantic masks into fine-grained dynamic confidence representation, and delivers the normalized gating map to the pose optimization stage for feature weighting and dynamic suppression.

(d) Dynamic-Aware Pose Optimization. Based on the gating map provided by DGCM and the front-end initial pose, this module first divides keypoints into dynamic and static subsets, and calculates the dynamic feature rejection ratio in the current frame. Then it adaptively adjusts the Huber robust kernel threshold according to the dynamic proportion, and finally conducts non-linear optimization on the $SE(3)$ manifold to refine camera pose. Only high-confidence static keypoints are involved in the optimization, which fundamentally suppresses the pose drift caused by moving objects.

This hierarchical design allows TP-DSLAM to maintain real-time performance while achieving superior robustness against dynamic disturbances.

3.2 Temporal-Latent Semantic Propagation (TLSP)

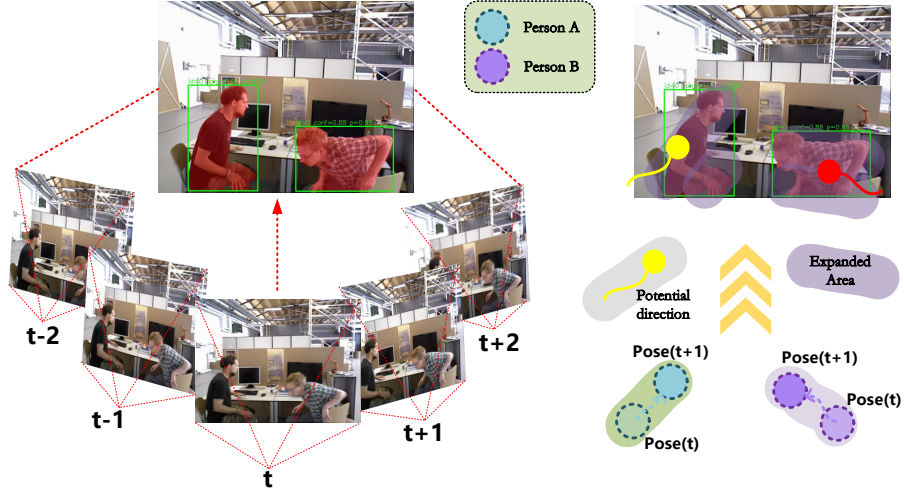


Fig. 2. Illustration of Temporal-Latent Semantic Propagation (TLSP). The pipeline follows a sequential order: pose estimation first, then semantic prior warping, and finally potential region expansion to handle motion uncertainty.

To alleviate the computational bottleneck of frame-by-frame dense semantic inference, we propose the Temporal-Latent Semantic Propagation (TLSP) strategy. As illustrated in Fig. 2, our framework avoids running heavy segmentation networks on every frame. Instead, it performs sparse semantic inference at key intervals and propagates the obtained priors to intermediate frames by exploiting temporal scene correlation.

We model semantic propagation as a recursive updating process. Let $\Psi(I_t)$ denote the semantic inference function. The network is activated only every K frames to balance localization accuracy and efficiency. For intermediate frames, the semantic prior map $\mathcal{P}_t$ is updated by warping the previous prior $\mathcal{P}_{t-1}$ with the estimated relative pose $\hat{T}_{t,t-1}$:

$$\mathcal{P}_t(u, v) = \gamma_t \cdot \Psi(I_t)(u, v) + (1 - \gamma_t) \cdot \mathcal{T}_{\text{warp}}(\mathcal{P}_{t-1}, \hat{T}_{t-1,t})(u, v) \quad (1)$$

The warping operator $\mathcal{T}_{\text{warp}}$ projects the prior semantic information onto the current image plane using the relative pose $\hat{T}_{t,t-1} \in SE(3)$. Following [10], this mapping is formulated as a differentiable sampling process:

$$\mathbf{P}_t^{\text{warp}} = \mathcal{B}\left(\mathbf{P}_{t-1}, \pi\left(\hat{T}_{t,t-1} \cdot \mathcal{M}(\mathbf{D}_t)\right)\right) \quad (2)$$

where $\pi(\cdot)$ is the camera projection function, $\mathcal{M}(\cdot)$ denotes the back-projection operation, $\mathbf{D}_t$ is the depth map, and $\mathcal{B}(\cdot)$ implements bilinear interpolation for sub-pixel accuracy.

To compensate for motion uncertainty and incomplete object boundaries, we apply a morphological dilation operation on the warped semantic prior:

$$\mathbf{P}_t^{\text{final}} = \mathbf{P}_t^{\text{warp}} \oplus \mathcal{K} \quad (3)$$

where $\mathcal{K}$ is the dilation structural kernel. The dilated prior mask is then fed into the DGCM module to ensure full coverage of potential dynamic regions.

As shown in Fig. 2, the pose-guided warping scheme reliably predicts dynamic object locations across frames. With mask dilation, our method effectively compensates for fast motion, suppresses boundary leakage, and maintains robustness in highly dynamic scenarios.

3.3 Disparity-Gated Confidence Modulation (DGCM)

Motivated by the widely recognized stereo SLAM principle that disparity reliability decays with distance [7, 15], we propose a disparity-gated confidence modulation (DGCM) mechanism. This mechanism addresses the limited generalization capability of pure semantic detectors by embedding binocular geometric constraints into the probabilistic space, using a sigmoid-based gating function to modulate semantic confidence according to normalized disparity.

For a keypoint k_i , its observed disparity d_i is assumed to follow a Gaussian distribution with mean μ_d and variance σ_d^2 , which is determined by stereo matching uncertainty. We define a **Geometric Credibility Gating Function** $\mathcal{G}(d_i)$ as:

$$\mathcal{G}(d_i; \theta) = \sigma\left(\beta \cdot \log\left(\frac{d_i}{d_0}\right) + \eta\right) \quad (4)$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, which maps the output to the interval $(0, 1)$; d_0 is the **characteristic depth baseline** of the scene, defined as the median disparity of the current frame; β is the **depth sensitivity**

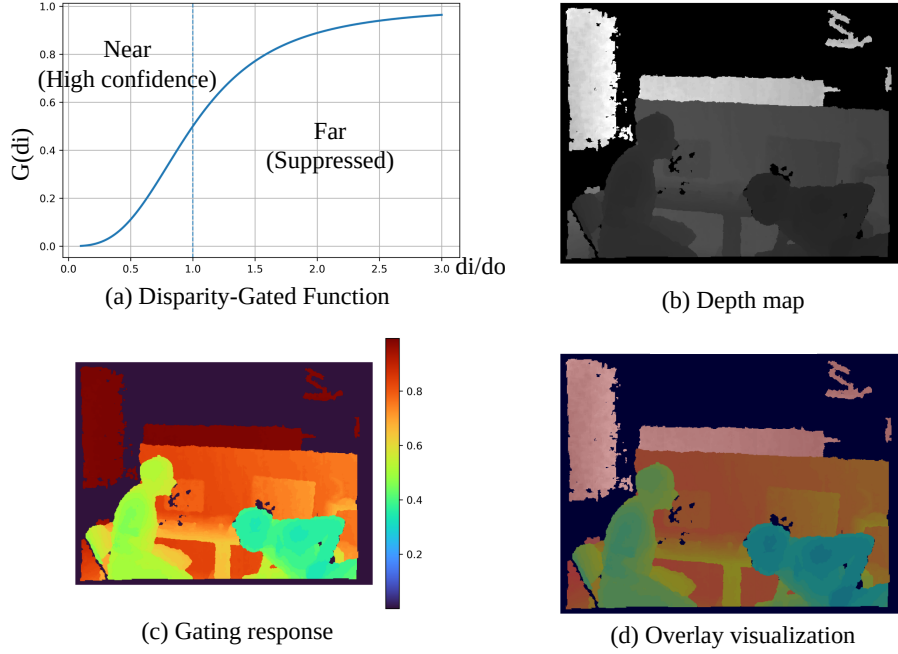


Fig. 3. Illustration of the proposed Disparity-Gated Confidence Modulation (DGCM). (a) The sigmoid-based gating function with respect to normalized disparity. (b) The input depth map. (c) The generated gating response, highlighting near-field regions. (d) Overlay visualization of the modulated confidence map on the original scene.

coefficient, which controls the suppression strength in near-range regions; and η is a bias term used to balance the global false-positive rate.

The physical interpretation of this function is that it constructs a nonlinear depth-aware manifold. When $d_i \gg d_0$ (near field), $\mathcal{G}(d_i) \rightarrow 1$, indicating that the geometric evidence strongly supports the hypothesis that, if a semantically dynamic feature exists in this region, it is highly likely to be truly dynamic. In contrast, when $d_i \ll d_0$ (far field), $\mathcal{G}(d_i) \rightarrow 0$, meaning that the geometric evidence becomes weaker, and the system is more inclined to rely on the smoothness of the semantic prior.

Finally, the **Joint Dynamic Posterior Score** of point k_i , denoted by $\mathcal{S}_i$, is defined as a nonlinear transformation of the semantic likelihood and geometric gating:

$$\mathcal{S}_i = \mathcal{F}_{\text{fusion}}(\mathcal{P}_t(u_i, v_i), \mathcal{G}(d_i; \theta)) = (\mathcal{P}_t(u_i, v_i))^{\lambda_1} \cdot (1 + \mathcal{G}(d_i; \theta))^{\lambda_2} \quad (5)$$

where λ_1 and λ_2 are learnable hyperparameters, or balancing factors determined via grid search.

The decision rule is formulated as a hypothesis test where the label z_i is assigned to 1 (Dynamic) if the posterior score $\mathcal{S}_i$ exceeds the optimal threshold τ_{opt} , and 0 (Static) otherwise. Here, τ_{opt} denotes the decision boundary obtained by minimizing the trajectory drift error on the validation set.

3.4 Dynamic-Aware Pose Optimization

The dynamic posterior score S_i yielded by the DGCM module quantifies the dynamic probability of each keypoint, providing a reliable basis for subsequent pose refinement. Instead of discarding dynamic features with binary masking, this work integrates the probabilistic confidence into geometric optimization via soft feature weighting.

A continuous weighting function is constructed to map the dynamic likelihood to optimization contribution:

$$\omega(S_i) = \exp(-\alpha \cdot S_i), \quad (6)$$

where α denotes a scaling factor. Keypoints with negligible dynamic probability maintain full weights during optimization, whereas those with high dynamic likelihood are adaptively suppressed.

The overall camera pose is solved by minimizing the weighted reprojection error over all keypoints on the $SE(3)$ manifold:

$$T_{tw}^* = \arg \min_{T \in SE(3)} \sum_{i \in \mathcal{K}_t} \omega(S_i) \cdot \rho_{\text{DA}} \left(\|\pi(T \cdot X_i) - z_i\|_{\Sigma_i}^2 \right) \quad (7)$$

where $\pi(\cdot)$ is the projection operation, X_i and z_i represent 3D map points and corresponding 2D keypoint observations, respectively, and Σ_i denotes the feature covariance matrix.

To adapt the robustness level to varying dynamic intensities, the Huber threshold is adjusted according to the global dynamic ratio η_{drop} , which statistically reflects the proportion of dynamic keypoints in the current frame. The threshold evolves following an exponential decay formulation:

$$\delta(\eta_{\text{drop}}) = \delta_0 \cdot e^{-\kappa \cdot \eta_{\text{drop}}} \quad (8)$$

where δ_0 is the baseline Huber threshold under static scenes, and κ is a pre-defined modulation coefficient that governs the decay rate of the truncation threshold. Larger κ leads to more aggressive threshold reduction in highly dynamic scenarios, thereby strengthening outlier suppression.

Remark 1. The proposed optimization mechanism realizes dual robustness regulation from individual feature confidence and global scene dynamics. The weighting function suppresses unreliable keypoints at the local level, while the adaptive Huber threshold automatically tunes the optimization sensitivity according to scene complexity, effectively mitigating pose drift in dynamic environments.

4 Experiments

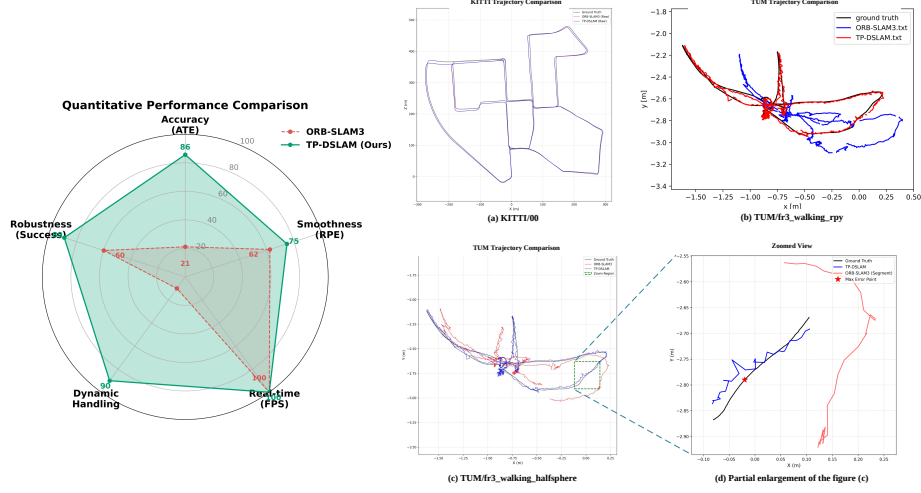


Fig. 4. The radar chart (left) quantitatively compares TP-DSLAM and ORB-SLAM3 across five metrics, where real-time capability is scored as full marks if requirements are met, and accuracy is evaluated via ATE and RPE on multiple datasets. The trajectory plots (right) visually demonstrate the superior tracking precision and robustness of our method compared to the ground truth.

This section evaluates the performance of the proposed TP-DSLAM system on multiple standard benchmark datasets. We first introduce the datasets and experimental setup. We then compare TP-DSLAM with state-of-the-art methods to validate its robustness in dynamic indoor scenarios, large-scale outdoor scenes and extreme visual conditions. Finally, ablation studies are performed to verify the effectiveness of each proposed component. All experiments are implemented on a laptop with an 13th Gen Intel(R) Core(TM) i5-13500H (2.60 GHz) processor to guarantee fair and consistent evaluation.

4.1 Dataset and Evaluation Metrics

To comprehensively assess the system, we utilize four widely adopted benchmark datasets: TUM RGB-D, KITTI, EuRoC, and VIODE.

TUM RGB-D Dataset: Used to evaluate tracking accuracy in dynamic indoor environments. It contains sequences with both static backgrounds and heavy dynamic interference.

KITTI Dataset: Used to assess scalability in large-scale outdoor autonomous driving scenarios.

EuRoC Dataset: Contains sequences characterized by fast motion and aggressive lighting changes, posing challenges to feature matching.

VIODE Dataset: Specifically designed to test performance under adverse visual conditions, such as day-night transitions and low-light environments.

We adopt the Absolute Trajectory Error (ATE RMSE) as the primary metric for quantitative assessment, which measures the difference between the estimated trajectory and the ground truth.

4.2 Comparison with State-of-the-Art Methods

We compare TP-DSLAM against several state-of-the-art approaches, including ORB-SLAM3 [4], DynaSLAM [2], and Dyna-VINS [13].

Table 1. Results on TUM dataset(RMSE in meters)

Sequences	ORB-SLAM3	YDD-SLAM	DynaSLAM	Dyna-VINS	Ours
w_half	0.312	0.028	0.025	0.061	0.026
w_xyz	0.510	0.015	0.015	0.049	0.015
w_rpy	0.536	0.036	0.035	0.063	0.035
w_static	0.164	0.008	0.006	0.008	0.009
s_half	0.024	×	0.017	×	0.019
s_xyz	0.009	×	0.015	×	0.013
Avg.(w)	0.381	0.022	0.020	0.045	0.021

Performance on TUM RGB-D Dataset ORB-SLAM3 suffers from obvious drift in highly dynamic TUM sequences due to the static-world assumption. Our TP-DSLAM effectively alleviates this problem, reducing the RMSE of the w_half sequence from 0.312 m to 0.026 m. On dynamic TUM scenes, our average error reaches 0.021 m, which is much lower than 0.381 m of ORB-SLAM3 and comparable to YDD-SLAM (0.022 m) and DynaSLAM (0.020 m). Moreover, TP-DSLAM maintains high precision of 0.009 m on static sequences, confirming its good generalization ability.

Performance on KITTI Dataset In challenging outdoor KITTI scenarios with moving vehicles and scale drift, TP-DSLAM achieves comparable localization accuracy with ORB-SLAM3 on sequences 00, 02 and 03, and clearly suppresses dynamic drift on the highly dynamic sequence 01. Although DynaSLAM II performs slightly better in some mapping tasks, our method still obtains competitive overall accuracy.

Table 2. Results on KITTI

Seq.	ORB3	DS(II)	DS	Ours
00	1.19	1.29	1.35	<u>1.20</u>
01	14.33	2.31	<u>2.42</u>	3.41
02	5.47	0.91	<u>1.04</u>	5.16
03	1.39	0.69	<u>0.78</u>	1.27
04	<u>0.25</u>	1.42	1.52	0.24
05	<u>0.91</u>	1.34	1.22	0.89
06	0.98	0.19	0.19	<u>0.92</u>
07	<u>0.44</u>	3.10	2.69	0.41
08	3.49	<u>1.68</u>	1.29	3.27
09	1.96	5.02	3.55	<u>1.98</u>
10	1.45	<u>1.30</u>	1.84	1.17

Table 3. Results on EuRoC

Seq.	ORB3	Ours	Imp.(%)
MH01	0.037	0.034	8.11
MH02	0.025	0.017	32.00
MH03	0.048	0.034	29.17
MH04	0.087	0.062	28.74
MH05	0.061	0.054	11.48
V101	0.030	0.029	3.33
V102	0.062	0.060	3.33
V103	0.056	0.054	3.57
V201	0.071	0.069	2.82
V202	0.061	0.059	3.29
V203	0.233	0.077	66.95

Performance on Challenging Conditions (EuRoC & VIODE) We further evaluate the system’s capability under fast motion and extreme lighting variations using the EuRoC and VIODE datasets.

EuRoC Dataset: As shown in Table 3, TP-DSLAM achieves prominent accuracy gains. In the challenging V203 sequence with severe illumination variation and motion blur, our method reduces the RMSE from 0.233 m to 0.077 m, bringing a performance improvement of 66.95

VIODE Dataset: As shown in Table 4.2, ORB-SLAM3 degrades severely under low-light conditions. By contrast, TP-DSLAM maintains stable accuracy and outperforms baseline methods in dynamic and low-light scenes such as City_night and Parking_lot. For extremely harsh cases like City_day_High and Parking_lot_High, DynaVINS with visual-inertial fusion performs better, implying that integrating inertial information can further improve the robustness of our method.

Table 4. Performance comparison under different scenarios (Spacious)

Method	City_day			City_night			Parking_lot		
	low	Mid	High	low	Mid	High	low	Mid	High
ORB-SLAM3	0.198	0.246	12.218	0.33	1.031	2.705	0.043	5.950	7.741
DynaSLAM	1.426	1.638	×	3.314	3.074	3.865	0.170	×	×
DynaVINS[13]	0.178	0.091	0.148	<u>0.182</u>	0.201	0.198	<u>0.042</u>	0.064	0.042
Ours	0.126	<u>0.128</u>	7.098	0.156	0.470	0.477	0.020	5.656	5.992

4.3 Ablation Study

To validate the contribution of individual components within the TP-DSLAM framework, we conduct systematic ablation experiments on the TUM dataset (specifically the *w_half* sequence). We evaluate the impact of our proposed modules: the Temporal Fusion Strategy (Time-Semantic Cache), the Geometric Gating mechanism (Disparity Gating), and the Dynamic Confidence weighting strategy.

Table 5. Ablation study on TUM *w_half*.

Method	RMSE (m) FPS	
ORB-SLAM3	0.312	118
w/o Temp. Fusion	0.031	12
w/o Geo. Gating	0.051	32
Hard-Rejection	0.032	34
Ours	0.026	30

Table 6. Runtime on Jetson Orin NX.

Method	Time (ms) FPS	
ORB-SLAM3	57	17
TP-DSLAM (Ours)	77	13

Note: RMSE refers to ATE RMSE ↓; Time (ms) indicates Processing Time per frame.

The results, summarized in Table 5, demonstrate that each component plays a vital role in achieving optimal performance.

w/o Temporal Fusion: Removing the temporal cache leads to a significant increase in error (0.031 m), indicating that historical semantic context is crucial for handling occlusions and dynamic noise.

w/o Geometric Gating: Disabling the disparity-based gating mechanism results in the highest error increase among the variants (0.051 m), highlighting its importance in filtering outliers.

Hard-Rejection: Using a simple hard-rejection strategy (instead of soft weighting) yields an error of 0.032 m, proving that our proposed dynamic confidence weighting preserves more useful features than binary masking.

Our full method achieves the lowest ATE of **0.026 m**, confirming the synergistic effect of the proposed modules.

4.4 Edge Device Deployment Evaluation

To validate real-world edge deployability, we benchmark TP-DSLAM against ORB-SLAM3 on an **NVIDIA Jetson Orin NX**. The system processes 640×480 stereo sequences, utilizing **YOLOv8-n** accelerated via CUDA for efficient semantic detection.

Runtime Performance Table 6 summarizes the runtime performance comparison on the Jetson Orin NX platform.

The results demonstrate that TP-DSLAM achieves real-time performance on the embedded platform, maintaining XX FPS while incorporating dynamic object filtering capabilities. Compared to ORB-SLAM3, our method introduces additional computational overhead from the TLSP and DGCM modules, yet remains within the real-time constraint for practical applications.

Accuracy-Efficiency Trade-off To analyze the trade-off between localization accuracy and processing speed, we evaluate different TLSP propagation intervals K on the TUM *w_half* sequence. Table 7 presents the results.

Table 7. Impact of TLSP interval K on accuracy and efficiency (Jetson Orin NX).

K	ATE RMSE (m)	FPS	Semantic Inference (ms)
1 (every frame)	0.012	7	35
3	0.013	10	27
5	0.015	13	25
10	0.022	17	20

Increasing K improves FPS by reducing computational load but risks degrading accuracy due to semantic staleness. We select $K = 5$ to balance efficiency and precision.

5 Conclusion

This paper presents TP-DSLAM, a robust stereo Visual SLAM framework designed for dynamic environments. The proposed system addresses the critical trade-off between computational efficiency and accuracy in dynamic scene understanding through two key innovations: Temporal-Latent Semantic Propagation (TLSP) and Disparity-Gated Confidence Modulation (DGCM).

TLSP reduces computational overhead and maintains semantic consistency via sparse inference and temporal warping, ensuring real-time edge performance. Concurrently, DGCM fuses binocular geometry with semantic likelihood for fine-grained dynamic suppression, substantially outperforming traditional hard-rejection strategies. Extensive experiments across multiple benchmark datasets validate the effectiveness of TP-DSLAM. The system achieves competitive performance with state-of-the-art methods while maintaining real-time operation.

However, like any research work, TP-DSLAM has certain limitations. The system’s performance in extreme conditions (e.g., very low light or severe motion blur) could be further improved. Additionally, the current implementation relies on stereo input, which may limit its applicability in monocular scenarios. Future work will focus on: exploring adaptive mechanisms for the TLSP propagation interval K based on scene dynamics; and incorporating loop closure detection with dynamic awareness to improve long-term mapping consistency.

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